

Some simple JavaScript examples

1. JavaScript links



```
<!DOCTYPE html>
<html>
  <head>
    <meta charset="utf-8">
    <title>JavaScript links</title>
    <style>
      body {
        margin: 40px; background-color: rgb(200,199,200);
      }
      div.link {
        margin-left: 30px;
      }
      h1 {
        font: Verdana, sans-serif medium navy;
      }
      a {
        color: red;
      }
    </style>

    <script>
      function showMessage(i) {
        alert("You have clicked the link n. " + i);
      }
    </script>
  </head>
  <body>
    <h1>Click a link...</h1>
    <div class="link">
      <a href="javascript:showMessage(1)">Link n. 1</a><br>
      <a href="javascript:showMessage(2)">Link n. 2</a><br>
      <a href="javascript:showMessage(3)">Link n. 3</a><br>
      <a href="javascript:showMessage(4)">Link n. 4</a><br>
      <a href="javascript:showMessage(5)">Link n. 5</a>
    </div>
  </body>
</html>
```

2. Dynamic content generation – Date and Time (version 1)

Date and time below are dynamically generated when the page is loaded

14/12/2021, 17:27:18

If you [reload](#) the page, the time will change

```
<!DOCTYPE html>
<html>
  <head>
    <meta charset="utf-8">
    <title>Dynamic content</title>
    <style>
      body {
        margin: 30px;
        background-color: rgb(250,238,133);
      }
      h1 {
        font-family: serif; font-size: normal;
        color: navy;
      }
    </style>
  </head>
  <body>
    <h1>
      Date and time below are dynamically<br>
      generated when the page is loaded
    </h1>
    <script>
      var today = new Date();
      document.write("<h2>" + today.getDate() + "/" +
        (today.getMonth() + 1) + "/" +
        today.getFullYear() + ", " +
        today.getHours() + ":" +
        today.getMinutes() + ":" +
        today.getSeconds() + "</h2>");
    </script>
    <p>
      If you <a href="javascript:document.location.reload()">
      reload</a> the page, the time will change
    </p>
  </body>
</html>
```

3. Dynamic content generation – Date and Time (version 2)

Date and time below are dynamically generated

Date: 14/12/2021

Time: 17:31:10

```
<!DOCTYPE html>
<html>
  <head>
    <meta charset="utf-8">
    <title>Dynamic content</title>
    <style>
      body {
        margin: 30px;
        background-color: rgb(250,238,133);
      }

      h1 {
        font-family: serif; font-size: normal;
        color: navy;
      }
    </style>
  </head>
  <body>
    <h1>
      Date and time below are dynamically<br>generated
    </h1>
    <script>
      var today = new Date();
      document.write("<h2>Date: <em>" + today.getDate() + "/" +
                    (today.getMonth() + 1) + "/" +
                    today.getFullYear() + "</em></h2>");
    </script>
    <p id="curtime"></p>
    <script>
      function checkZero(num) {
        return ('0' + num).slice(-2);
      }

      function showTime() {
        var t = new Date();
        document.getElementById("curtime").innerHTML = "<h2>Time: " +
          checkZero(t.getHours()) + ":" +
          checkZero(t.getMinutes()) + ":" +
          checkZero(t.getSeconds()) + "</h2>";
      }

      showTime();
      setInterval(showTime, 1000);
    </script>
  </body>
</html>
```

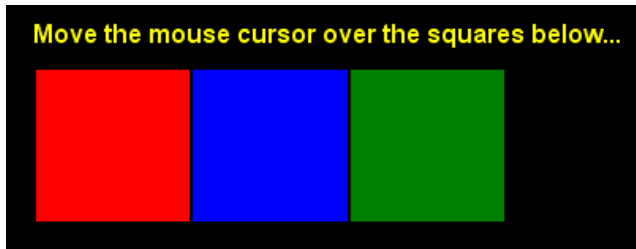
4. History links

Click the following links to go back and forward in the browser history

[Back -2](#) [Back -1](#) [Forward +1](#) [Forward +2](#)

```
<!DOCTYPE html>
<html>
  <head>
    <meta charset="utf-8">
    <title>History links</title>
    <style>
      body {
        margin: 30px;
        font-family: Arial, sans-serif;
      }
      h1 { font-size: 1.5em }
    </style>
  </head>
  <body>
    <h1>
      Click the following links to go back and forward in the browser history
    </h1>
    <a href="javascript:history.go(-2)">Back -2</a>&nbsp;&nbsp;&nbsp;
    <a href="javascript:history.go(-1)">Back -1</a>&nbsp;&nbsp;&nbsp;
    <a href="javascript:history.go(1)">Forward +1</a>&nbsp;&nbsp;&nbsp;
    <a href="javascript:history.go(2)">Forward +2</a>
  </body>
</html>
```

5. Rollover: background color change



```
<!DOCTYPE html>
<html>
  <head>
    <meta charset="utf-8">
    <title>Rollover background</title>
    <style>
      body {
        background: black;
        margin: 2em;
      }
      h1 {
        font-family: sans-serif;
        font-size: medium;
        color: yellow;
      }
    </style>
    <script>
      function bgChange(bg) {
        document.body.style.background = bg;
      }
    </script>
  </head>
  <body>
    <h1>Move the mouse cursor over the squares below...</h1>
    <table width="300" height="100">
      <tr>
        <td onmouseover="bgChange('red')" onmouseout="bgChange('black') "
          style="background-color: red">
        </td>
        <td onmouseover="bgChange('blue')" onmouseout="bgChange('black') "
          style="background-color: blue">
        </td>
        <td onmouseover="bgChange('green') " onmouseout="bgChange('black') "
          style="background-color: green">
        </td>
      </tr>
    </table>
  </body>
</html>
```

6. Rollover: image resizing

Move the mouse cursor over the image below...



```
<!DOCTYPE html>
<html>
  <head>
    <meta charset="utf-8">
    <title>Interactive image resizing</title>
    <script>
      function moveOver() {
        image.width = "210";
        image.height = "210";
      }

      function moveBack() {
        image.width = "105";
        image.height = "105";
      }
    </script>
  </head>
  <body>
    <h3>Move the mouse cursor over the image below...</h3>
    
  </body>
</html>
```

7. Rollover: image replacement

When the mouse cursor is moved over the link-image `im1.jpg`, the image is replaced with `im2.jpg`, and when the cursor is moved away from the image, `im1.jpg` is displayed again (effect also achievable through CSS)

```
<!DOCTYPE html>
<html>
  <head>
    <meta charset="utf-8">
    <title>A simple JavaScript rollover</title>
  </head>
  <body>
    <a href="http://www.google.com"
      onmouseover="document.getElementById('imroll').src='im2.jpg'"
      onmouseout="document.getElementById('imroll').src='im1.jpg'">
      
    </a>
  </body>
</html>
```

8. Interactive change of CSS properties



```
<!DOCTYPE html>
<html>
  <head>
    <meta charset="utf-8">
    <title>Interactive change of CSS properties</title>
    <style>
      #ta {
        background: #e6e6e6;
        font-family: Arial, sans-serif;
        font-size: 1em;
        float: left;
        margin-right: 30px;
        width: 350px;
        height: 180px;
      }
    </style>
    <script>
      var color = new Array(4);
      color[0] = "#e6e6e6";
      color[1] = "#ff5c1c";
      color[2] = "#c3fbc3";
      color[3] = "#99e3ff";

      var size = new Array(3);
      size[0] = "1em";
      size[1] = "1.5em";
      size[2] = "2em";

      function bgchange() {
        for (var i=0; i < document.f.bgc.length; i++) {
          if (document.f.bgc[i].checked) {
            document.getElementById("ta").style.backgroundColor = color[i];
          }
        }
      }

      function fschange() {
        for (var i=0; i < document.f.dim.length; i++) {
          if (document.f.dim[i].checked) {
            document.getElementById("ta").style.fontSize = size[i];
          }
        }
      }
    </script>
  </head>
```


9. Cm-In/In-Cm conversion

Two text input fields with conversion from centimeters to inches and vice versa (through the 'onchange' event handler)

Centimeters	38.1	Inches	15
Convert			

```
<!DOCTYPE html>
<html>
  <head>
    <meta charset="utf-8">
    <title>Conversions</title>
    <script>
      function cmToIn() {
        document.convform.inches.value = document.convform.cm.value / 2.54;
      }
      function inToCm() {
        document.convform.cm.value = document.convform.inches.value * 2.54;
      }
    </script>
  </head>
  <body>
    <form name="convform">
      <label for="cm">Centimeters</label>
      <input type="text" id="cm" size="15" onchange="cmToIn()">
      &nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&~
&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&~
      <label for="inches">Inches</label>
      <input type="text" id="inches" size="15" onchange="inToCm()">
      <br><br>
      <input type="button" value="Convert">
    </form>
  </body>
</html>
```

10. Rotating banners

```
<!DOCTYPE html>
<html>
  <head>
    <meta charset="utf-8">
    <title>Rotating banners</title>
    <script>
      // Indicates which image must be currently displayed (0, 1 or 2 if we have
      // three banners)
      cur_img = 0;

      // Let's create two arrays, one for images and one for the
      // corresponding URLs
      image = new Array(3);
      url = new Array(3);

      // Let's assign values to the arrays..
      image[0] = "wmt.png"; image[1] = "mpce.png"; image[2] = "unipv.png";
      url[0] = "https://vision.unipv.it/wmt/";
      url[1] = "https://mpce.unipv.eu/";
      url[2] = "https://portale.unipv.it/";

      // Each time it is executed, 'changeImage' displays the next image in the
      // sequence (first, second, third, first, second, ...)

      function changeImage() {
        cur_img++;
        if ( cur_img == 3 )
          cur_img = 0;
        // The "new" image is displayed
        document.getElementById("bannerimg").src = image[cur_img];
      }

      // 'loadPage' opens (in a new window named 'new') the page
      // associated to the image

      function loadPage() {
        open(url[cur_img], "new");
      }

      // 'setInterval' calls the specified function at regular time intervals
      setInterval(changeImage, 2000);
    </script>
  </head>

  <body>
    <!-- Initially, 'wmt.png' is displayed -->
    <a href="javascript:loadPage()">
      
    </a>
    <h1>Rotating banners</h1>
    <p>
      The image changes at regular time intervals. By clicking it, you'll
      load an associated web page
    </p>
  </body>
</html>
```

11. Factorial calculation

Factorial calculation

Number

```
<!DOCTYPE html>
<html>
  <head>
    <meta charset="utf-8">
    <title>Factorial calculation</title>
    <script>
      function fact(n) {
        f = 1;
        if (n < 0)
          return -1;
        i = 1;
        do {
          f = f * i;
          i = i + 1;
        } while (i <= n);
        return f;
      }

      function calculateFact(n) {
        f = fact(n);
        if (f == -1)
          alert("Error!");
        else
          alert("The factorial of " + n + " is " + f);
      }
    </script>
  </head>
  <body>
    <h1>Factorial calculation</h1>
    <form id="myform" name="myform">
      Number <input type="text" size="5" id="num">
      <input type="button" value="Calculate"
        onclick="calculateFact(document.myform.num.value)">
    </form>
  </body>
</html>
```